Title: MicroGoals – Secure Goal Management Backend API

TEAM ID:SWTID-2026-4423

TEAM SIZE: 2

TEAM MEMBERS:

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Objective:

To develop a secure backend API for goal management
To allow users to create, update, and delete goals
To store goal information safely in a database
To provide API endpoints for frontend integration
To demonstrate REST API development

Project Overview :

MicroGoals is a secure backend RESTful API developed using Node.js, Express.js, and MongoDB. It enables users to register, log in, and manage their personal goals through CRUD operations. The system uses JWT authentication and middleware to ensure data security and user-specific access control.

The system includes:

- **User Registration & Login**
- **JWT-based Authentication**
- **CRUD Operations for Goals**
- **Protected Routes**
- **Centralized Error Handling**

The main objective is to provide a secure and scalable backend system for digital goal management.

Project Instruction:

- **The project follows REST API principles.**
- **Authentication is handled using JWT.**
- **MongoDB is used as the database.**
- **APIs are tested using Thunder Client.**
- **The project follows MVC architecture.**

Pre-requisites

Before running the project, the following tools are required:
Node.js (v18 or higher)
Express.js
MongoDB (Local / Atlas)
VS Code
Thunder Client (Extension)
Basic knowledge of JavaScript & REST APIs

TECHNICAL ARCHITECTURE

Runtime: Node.js

Framework: Express.js

Database: MongoDB

Authentication: JWT

Architecture Pattern: MVC

ER DIAGRAM

Entities:

1.User

_id

name

email

password

2.Goal

_id

title

completed

userId (Reference to User)

Relationship:

One User → Many Goals

FEATURES

- User Registration
- Secure Login with JWT
- Create Goal
- Update Goal
- Delete Goal
- View User-specific Goals
- Protected Routes
- Centralized Error Handling

ROLES AND RESPONSIBILITIES

Role	Responsibility
User	Register, Login, Manage Goals
Developer	API Development & Testing

Conclusion

The MicroGoals project demonstrates how a secure backend API can be developed to manage personal goals. The system allows users to create, update, and delete goals efficiently.

The project also provides practical experience in backend development, REST API design, and database integration.